

30W AUDIO POWER AMPLIFIER

Using NE5534 And Darlington Transistors

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There are many different audio power amplifier ICs to choose from, but designing an audio power amplifier using operational amplifier (op-amp) and bipolar transistors is quite advantageous.

Compared to audio ICs, power transistors are easy to replace in case of damage and have relatively:

1. Very high power dissipation
2. Very high maximal working voltages and collector currents

3. High maximal working temperature

Here, we use an NE5534 operational amplifier and two Darlington transistors (TIP141 and TIP146) to produce 30W power.

NE5534 was selected because it has maximal working power supply up to $\pm 22V$, significant bandwidth, low total harmonic distortion (THD), high slew rate and output current more than $\pm 20mA$.

Circuit and working

Fig. 1 shows the power supply section. Fig. 2 shows the circuit of the audio power amplifier using NE5534 and power transistors.

In the audio power amplifier circuit, the audio signal is applied to CON5. Capacitor C12 blocks the DC input signal. Resistor R4 and capacitor C13 limit the frequency of the input signal. Resistor R5 (100-kilo-ohm) determines the input resistance of the amplifier. The amplifier's gain is determined by resistors R6 and R7 as follows:

$$\text{Gain} = 1 + R7/R6 = 1 + 30k/1k = 31$$

The amplifier's gain can be set to any appropriate value above 3, because NE5534 is stable at gains higher than 3. Resistors R8 through R10 reduce the total harmonic distortion of the amplifier. Capacitor C18 and resistor R13 increase the amplifier's stability and reduce THD. Darlington transistors T1 and T2 amplify the output current of NE5534.

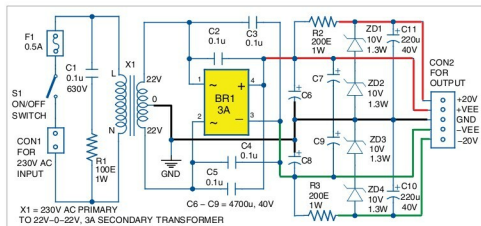


Fig. 1: Power supply circuit diagram

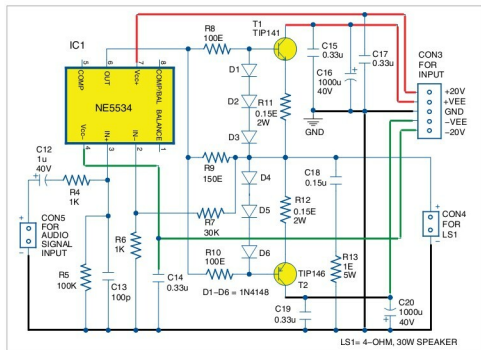


Fig. 2: Circuit diagram of the audio power amplifier